UNIVERSITY OF ENGINEERING AND TECHNOLOGY

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Assignment	**Lab Assignment 8**
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ASSIGNMENT 08

Q no: 1

Create a class named as base having protected integer data attribute and a function display(). Derive a class publically from base class which has function named as display(). Base and derive class display function should output a string as “I am base class display function” & “I am derived class display function” respectively. In main function create derived class object and call both base and derived class functions.

Lab8_1.cpp × Lab8_2.cpp Lab8_3.cpp Lab8_

Lab8_1.cpp > main()

```cpp
#include <iostream>
using namespace std;

class Base {
protected:
    int data;
public:
    void display() {
        cout << "I am base class display function" << endl;
    }
};

class Derived : public Base {
public:
    void display() {
        cout << "I am derived class display function" << endl;
    }
};

int main() {
    Derived obj;
    obj.Base::display();
    obj.display();
    return 0;
}
```

Output:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS E:\2nd Semester\1 OOPS\OOPS LAB\Lab Code> cd "e:\2nd Semeste
b8_1 }
I am base class display function
I am derived class display function
PS E:\2nd Semester\1 OOPS\OOPS LAB\Lab Code>
```

Q no: 2

Create a class named as base having protected integer data attribute as “a, b, c, d” and by default constructor, one argument constructor, two argument constructor, display() function which will print “I am base class display function” string and a destructor. Derive a class publically from base class which has own destructor as well. Create derive class object, one argument object and two argument object in main function. Call display function from derive class object. While creating derive class objects, the constructor of both base and derive class should display string of your own choice and should assign value to protected data attributes of base class.

Lab8_1.cpp | Lab8_2.cpp | Lab8_3.cpp | Lab8_

Lab8_2.cpp > base > base(int)

```cpp
#include <iostream>
using namespace std;

class base
{
protected:
    int a, b, c, d;
    public :
    base(){
        a=b=c=d=0;
        cout<<"I am the constructor of base class"<<endl;
    }
    base(int x){
        a=x; b=c=d=0;
        cout<<"I am the 1 argument constructor of base class, a ="<<a<<endl;

    }
    base(int z, int r){
        a=z,b=r; c=d=0;
    cout<<"I am the 2 argument constructor of base class a ="<<a<<" b ="<<b<<endl;

    }
```

```
Lab8_2.cpp > base > base(int)
class base
    base(int z, int r){
    void display()
    {
        cout << "== I am Base class display function ==" << endl;
        cout<<"Values are a ="<<a<<" b ="<<b<<" c ="<<c<<" d ="<<d<<endl;
    }
    ~base(){
        cout<<"Base destructor called"<<endl;

    }
};

class derived : public base
{
    public:
    derived(): base(){
    cout<<"== Derived default constructor called =="<<endl;
    }
    derived(int x):base(x){
       cout << "Derived one-argument constructor called." << endl;
    }
    derived(int x, int y):base(x,y){
        cout << "Derived two-argument constructor called." << endl;
    }
    ~derived(){
        cout << "== Derived destructor called ==" << endl;
    }
};
```

Output:

```
PROBLEMS 2    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS
PS E:\2nd Semester\1 OOPS\OOPS LAB\Lab Code> cd "e:\2nd Semester\1
b8_2 }
I am the constructor of base class
== Derived default constructor called ==
== I am Base class display function ==
Values are a =0 b =0 c =0 d =0
I am the 1 argument constructor of base class, a =10
Derived one-argument constructor called.
== I am Base class display function ==
Values are a =10 b =0 c =0 d =0
I am the 2 argument constructor of base class a =20 b =30
Derived two-argument constructor called.
== I am Base class display function ==
Values are a =20 b =30 c =0 d =0
== Derived destructor called ==
Base destructor called
== Derived destructor called ==
Base destructor called
== Derived destructor called ==
Base destructor called
PS E:\2nd Semester\1 OOPS\OOPS LAB\Lab Code>
```

Q no: 3

Write a C++ program that can read and print student information. The base class should have student's name, age and gender as private data members and two public member functions which will input student's private data member's information and another member function will print all the information. The derived class should be about student's result information and has private data members such as total marks, percentage and grade. The derived class has get and put member functions which will input private data member information and will display the student result information respectively. In the main function create derived class object and call both functions of the base class and derived class respectively

Lab8_1.cpp | Lab8_2.cpp | Lab8_3.cpp | Lab8_

Lab8_3.cpp > main()

```cpp
#include<iostream>
#include<string>
using namespace std;

class student_1{
   private:
   string name;
   int age;
   string gender;
   public:
   void getData(){
    cout<<"Enter name of the Student : ";cin>>name;
    cout<<"Enter age of the Student : ";cin>>age;
    cout<<"Enter gender of the Student : ";cin>>gender;
   }
   void display(){
    cout<<"Name of the Student is :"<<name<<endl;
    cout<<"Age of the Student is :"<<age<<endl;
    cout<<"Gender of the Student is :"<<gender<<endl;
   }
};
class student_2: public student_1{
    private:
    float totalMarks;
    float percentage;
    float grade;
```

```
    public:
    void get(){
        cout<<"Enter the total Marks : ";cin>>totalMarks;
        cout<<"Enter the Percentage : ";cin>>percentage;
        cout<<"Enter the Grade : ";cin>>grade;
    }
    void put(){
        cout<<"The total Marks are :"<<totalMarks<<endl;
        cout<<"Percentage is :"<<percentage<<endl;
        cout<<"Grades are :"<<grade<<endl;
    }
};
int main(){
    student_2 s1;
    s1.getData();
    s1.get();
    s1.display();
    s1.put();
    return 0;
}
```

Output:

```
PROBLEMS 2    OUTPUT    DEBUG CONSOLE    TERMINAL    POR

PS E:\2nd Semester\1 OOPS\OOPS LAB\Lab Code> cd "e:\
b8_3 }
Enter name of the Student : Asim
Enter age of the Student : 18
Enter gender of the Student : Male
Enter the total Marks : 94
Enter the Percentage : 90
Enter the Grade : 91.9
Name of the Student is :Asim
Age of the Student is :18
Gender of the Student is :Male
The total Marks are :94
Percentage is :90
Grades are :91.9
PS E:\2nd Semester\1 OOPS\OOPS LAB\Lab Code>
```

Q no 4:

Its from the given diagram

Lab8_1.cpp | Lab8_2.cpp | Lab8_3.cpp | Lab8_4.cpp

Lab8_4.cpp > ...

```cpp
#include <iostream>
using namespace std;

class Person {
public:
    void Talk() {
        cout << "Person is talking." << endl;
    }
    void Walk() {
        cout << "Person is walking." << endl;
    }
    void Eat() {
        cout << "Person is eating." << endl;
    }
};
class Mathsteacher : public Person {    "Mathsteacher": Unkn
public:
    void TeachMaths() {
        cout << "Teaching mathematics." << endl;
    }
};
class Footballer : public Person {
public:
    void PlayFootball() {
```

```
22 class Footballer : public Person {
23 public:
24     void PlayFootball() {
25         cout << "Playing football." << endl;
26     }
27 };
28 class Businessman : public Person {
29 public:
30     void RunBusiness() {
31         cout << "Running business." << endl;
32     }
33 };
34 int main() {
35     cout << "= Maths Teacher =" << endl;
36     Mathsteacher teacher;    "Mathsteacher": Unknown word.
37     teacher.Talk();
38     teacher.Walk();
39     teacher.Eat();
40     teacher.TeachMaths();
41
42     cout << "\n= Footballer =" << endl;
43     Footballer player;
44     player.Talk();
45     player.Walk();
46     player.Eat();
47     player.PlayFootball();
48
49     cout << "\n= Businessman =" << endl;
50     Businessman exec;
51     exec.Talk();
52     exec.Walk();
53     exec.Eat();
54     exec.RunBusiness();
55
56     return 0;
57 }
```

Output:

```
PROBLEMS 2    OUTPUT    DEBUG CONSOLE    TERMINAL    PORT

PS E:\2nd Semester\1 OOPS\OOPS LAB\Lab Code> cd "e:\2
b8_4 }
= Maths Teacher =
Person is talking.
Person is walking.
Person is eating.
Teaching mathematics.

= Footballer =
Person is talking.
Person is walking.
Person is eating.
Playing football.

= Businessman =
Person is talking.
Person is walking.
Person is eating.
Running business.
PS E:\2nd Semester\1 OOPS\OOPS LAB\Lab Code>
```

Q no:5

Imagine a publishing company that markets both book and audiocassette versions of its works. Create a class publication that stores the title (a string) and price (type float) of a publication. From this class derive two classes: book, which adds a page count (type int), and tape, which adds a playing time in minutes (type float). Each of these three classes should have a getdata() function to get its data from the user at the keyboard, and a putdata() function to display its data. Write a main() program to test the book and tape classes by creating instances of them, asking the user to fill in data with getdata(), and then displaying the data with putdata().

ASSIGNMENT 08

Lab8_5.cpp > ...

```cpp
#include <iostream>
#include <string>
using namespace std;


class Publication {
protected:
    string title;
    float price;
public:
    void getdata() {
        cout << "Enter title: ";
        getline(cin, title);
        cout << "Enter price: ";
        cin >> price;
        cin.ignore();
    }

    void putdata() const {
        cout << "Title: " << title << endl;
        cout << "Price: $" << price << endl;
    }
};
```

```cpp
class Book : public Publication {
private:
    int pageCount;
public:
    void getdata() {
        Publication::getdata(); // Get base class data
        cout << "Enter page count: ";
        cin >> pageCount;
        cin.ignore();
    }

    void putdata() const {
        Publication::putdata(); // Display base class data
        cout << "Page count: " << pageCount << " pages" << endl;
    }
};


class Tape : public Publication {
private:
    float playingTime;
public:
    void getdata() {
        Publication::getdata();
```

```cpp
        cout << "Enter playing time (in minutes): ";
        cin >> playingTime;
        cin.ignore();
    }

    void putdata() const {
        Publication::putdata();
        cout << "Playing time: " << playingTime << " minutes" << endl;
    }
};

int main() {
    Book myBook;
    Tape myTape;

    cout << "=== Enter Book Details ===" << endl;
    myBook.getdata();

    cout << "\n=== Enter Tape Details ===" << endl;
    myTape.getdata();

    cout << "\n=== Book Information ===" << endl;
    myBook.putdata();

    cout << "\n=== Tape Information ===" << endl;
```

```
    cout << "=== Enter Book Details ===" << endl;
    myBook.getdata();

    cout << "\n=== Enter Tape Details ===" << endl;
    myTape.getdata();

    cout << "\n=== Book Information ===" << endl;
    myBook.putdata();

    cout << "\n=== Tape Information ===" << endl;
    myTape.putdata();

    return 0;
}
```

Output:

```
PROBLEMS 2    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\2nd Semester\1 OOPS\OOPS LAB\Lab Code> cd "e:\2nd S
b8_5 }
=== Enter Book Details ===
Enter title: Pursuit of philosophy
Enter price: 200
Enter page count: 700

=== Enter Tape Details ===
Enter title: Pursuit of nowhere
Enter price: 1200
Enter playing time (in minutes): 1110

=== Book Information ===
Title: Pursuit of philosophy
Price: $200
Page count: 700 pages

=== Tape Information ===
Title: Pursuit of nowhere
Price: $1200
Playing time: 1110 minutes
PS E:\2nd Semester\1 OOPS\OOPS LAB\Lab Code>
```

The End